INFERRING COVID-19 SENTIMENTS OF CHINESE MICRO-BLOGGERS USING LLMS

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Abstract

Studying public sentiment during crises like the COVID-19 pandemic is crucial for understanding how opinions and sentiments shift and are polarized in society. Weibo, the most popular microblogging site in China, is a beautiful outlet for this study. Analyzing sentiment shifts on Weibo provides insights into how social events and government actions influence public opinion. This study contributes to understanding the dynamics of social sentiments during health crises, fulfilling a gap in sentiment analysis for Chinese platforms. By examining these dynamics, we aim to offer valuable perspectives on digital communication's role in shaping society's responses during unprecedented global challenges.

This study uses Large Language Models to analyze users' sentiments on the Weibo platform during the Covid-19 pandemic. The period includes the pre-COVID-19 period, the outbreak period, and the early stage of epidemic prevention. Focusing on how social events and government prevention measures (as external factors) and users' comments and retweets (as internal factors) influence sentiment. This research extracts the hidden patterns from the Weibo data, identifies complex social sentiments, and, in the end, predicts their dynamics across the internet. The study uses the Weibo COVID-19 Dataset from Harvard, classifying sentiments into positive, negative, sarcastic, and neutral categories. The goal is to examine the patterns of shifts in overall Weibo sentiments and individual sentiments over time and compare these patterns with the results of our predictions.

Dataset Introduction

The dataset is structured in JSON format and contains information such as the content of the post, repost details, timestamps, user information, and unique identifiers for both users and posts. Below is an example of the dataset structure:


```
{
    "From": "iPhone 14",
    "content": "Example Post",
    "repost": {
        "content": "Example Repost",
        "imgs": [],
        "timestamp": "2020-03-11T19:05",
        "username": 6604880503
    },
    "timestamp": "2020-03-12T01:40",
    "user_id": 7816496155,
    "weibo_id": "4481466803361052"
}
```


Fig. 1

The dataset contains a total of 4,049,407 posts. If we focus on the content of the posts, the dataset includes 2,322,148 distinct posts. The remaining 1,727,259 posts are duplicated content that may be simple reposts without any comments, human-made propaganda, or bot-generated spam:

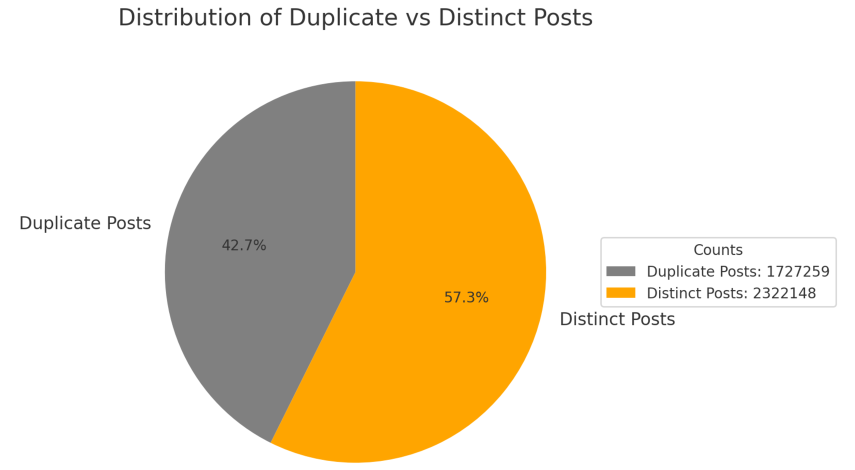


Fig. 2

Methodology

Cleaning Dataset:

As mentioned earlier, the dataset contains a significant number of duplicate posts. Some of these duplicates are straightforward to identify as they are entirely identical. However, certain spam posts employ subtle variations, such as the addition of spaces or tabs, to avoid spam filters on Weibo. To address this issue, we developed a program to preprocess and clean the Weibo dataset, ensuring its quality and consistency before proceeding with classification.

Posts Types and Counts

	Sentiments	Counts
1	Simple Reposts	945709
2	Original Posts	2226667
3	Duplicated Posts	877031

Fig. 3

Classification:

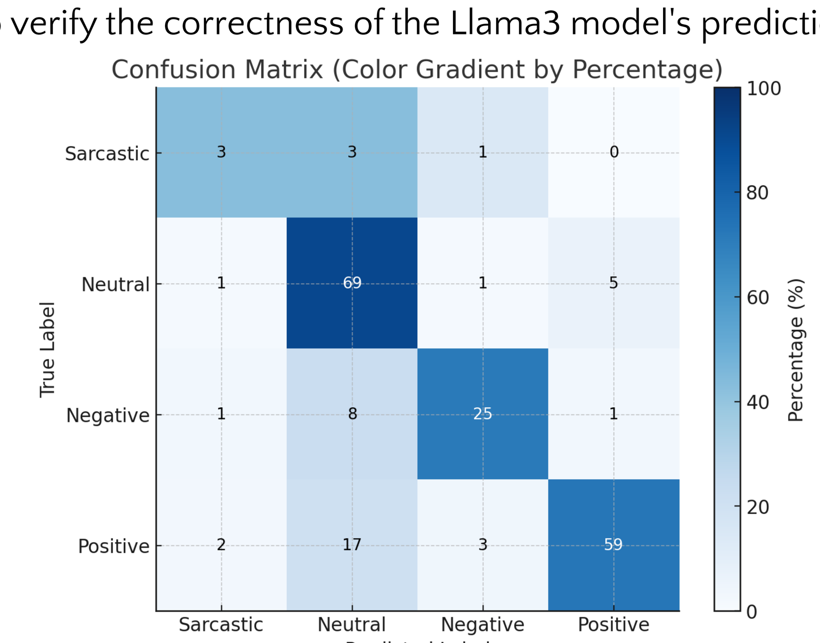
Sentiments are classified into four categories: positive, negative, sarcastic, and neutral. For sentiment labeling, the Llama 3 model running on the RPI supercomputer is used. The key to data labeling was the adoption of few-shot learning. We manually identified four posts containing each of the four sentiment types, then created a prompt for the Llama 3 model to predict the rest of the Weibo content based on the four manually labeled posts it had just learned. Below is the prompt:

```
You are a bot designed to judge Chinese sentences as having positive, negative or neutral sentiment. I'm going to provide posts from a Chinese social media. IMPORTANT: You must only answer with "positive", "negative", "sarcastic" or "neutral", using the following criteria for your judgement:
1. If the post speaks well of a figure or event that is commonly regarded as a good figure or event, with no unnecessary exaggeration, it is 'positive' sentiment.
2. If the post disparages a figure or event that is commonly regarded as a bad figure or event, then it is a 'negative' sentiment.
2. If the post has no emotional language and consists of matter-of-fact reporting of factual statements, that corresponds to a 'neutral' sentiment.
3. Sarcasm in the post, which appears in the form of exaggerated emotion, pretend naivete, or other common sarcastic tone indicators, should correspond to a 'sarcastic' sentiment.

IMPORTANT: Some posts may be appended with the original post that the user replied to. This will follow the format: [reply]//[original post]. You must predict the sentiment of the reply only, but you may use the original post as context to understand the reply.

Here are some examples:
Post: 1月3日李文亮医生在武汉公安局签署训诫书不要造谣接着他回到第一线继续接诊新冠病毒肺炎患者1月10号他开始出现症状随后住院此期间直到1月15日武汉卫建委对全国发出公告称尚未发现明确的人传人证据不能排除有限人传人的可能但持续人传人的风险较低说没发现有医护人员感染
Expected Answer: Neutral

Post: 妈妈 我已经快二十天没喝奶茶 没有大吃大喝了 求求你赶紧疫情结束 我人都快没了 我想上课我想喝奶茶我想吃烧烤
Expected Answer: Negative

Post: 高中时就暗恋他已久，高三毕业那天我鼓起勇气表白，万万没想到他居然也默默喜欢着我，填志愿时也选择了同一个城市。后来工作了异地了四年，晃眼我们走过了十年呢，19年时我们步入了婚姻礼堂。我想 这世上最幸福的事情之一 那就是两个人都互相深爱并且坚持吧~疫情过后 春暖花开，我们想去武大看樱花。
Expected Answer: Positive

Post: 就你们敢说实话，//@7362410961:世卫组织说目前只有瑞德西韦可能有效，中科院双黄连有效，南京大学说金银花有效，北京大学沐舒坦有效，南开大学说姜、大枣、龙眼肉都能预防。中国人民真幸福，这么多常见药物、食品可以抗新型冠状病毒，还慌什么呢?
Expected Answer: Sarcastic
```

Fig. 4

Validation:

Two hundred manually curated posts were used for validation. These posts were also processed through ChatGPT to serve as a reference, ensuring the accuracy and reliability of the validation set. This approach aims to verify the correctness of the Llama3 model's predictions.



Results

The total sentiment distribution is summarized as follows. Two types of visualizations were created: one chart illustrates the sentiment distribution across all posts, while the other focuses exclusively on distinct posts:

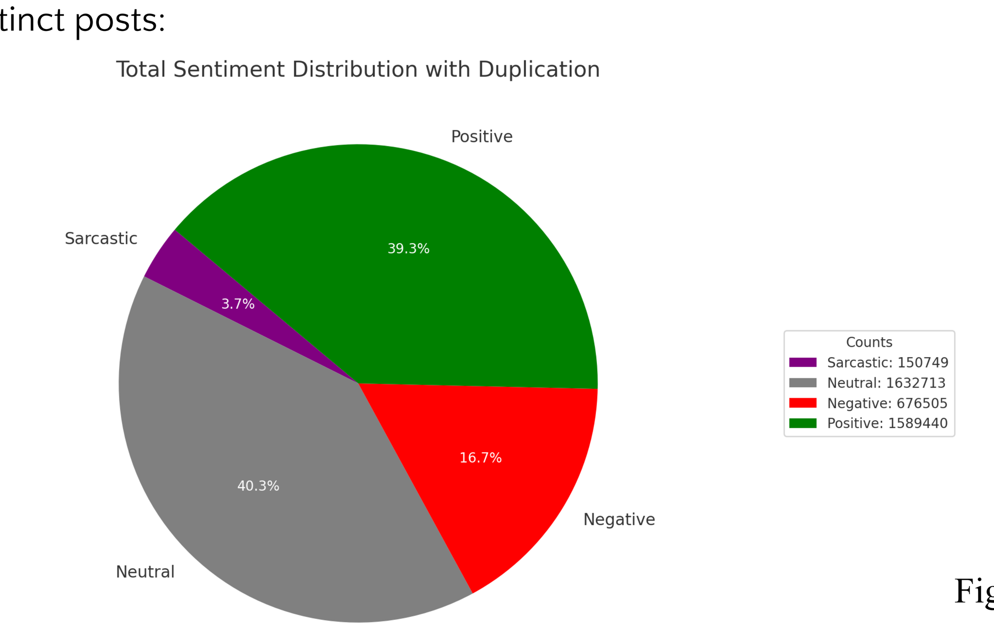


Fig. 6 a

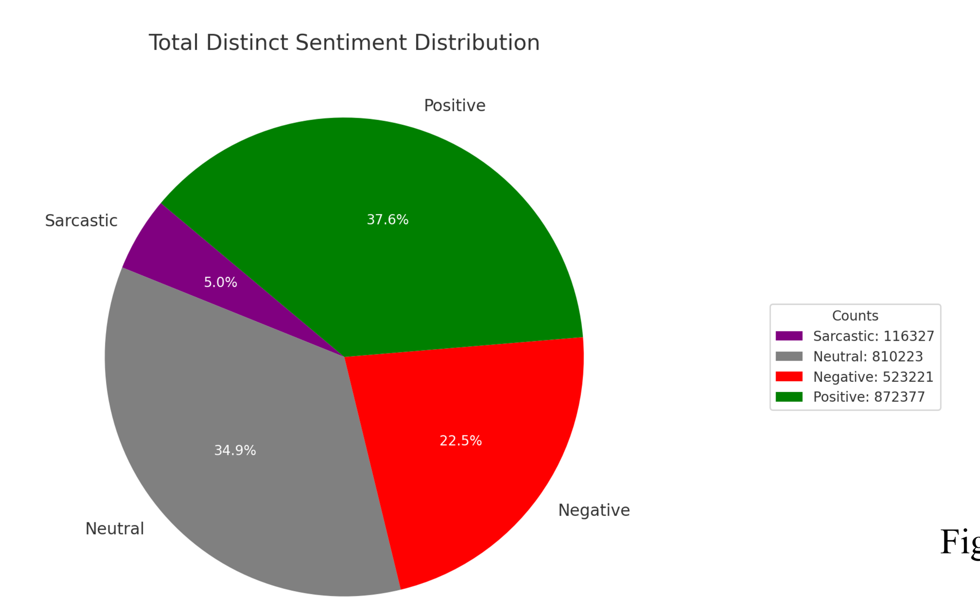


Fig. 6 b

We can also examine the evolution of sentiment dynamics over time. As with the previous analysis, two types of visualizations are provided: the first chart represents the sentiment dynamics for all posts, capturing the overall trends, while the second chart focuses on distinct posts, highlighting the sentiment changes for distinct posts over time.

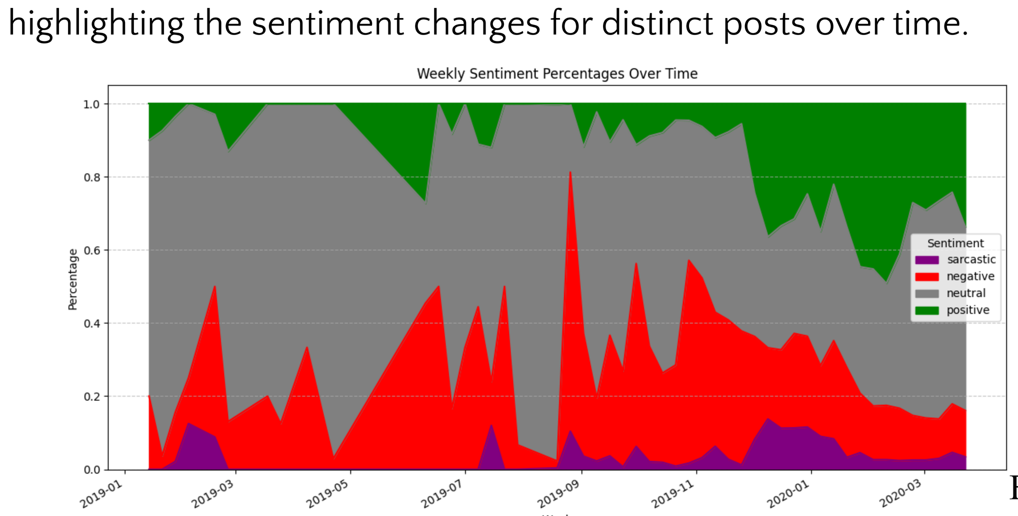


Fig. 7 a

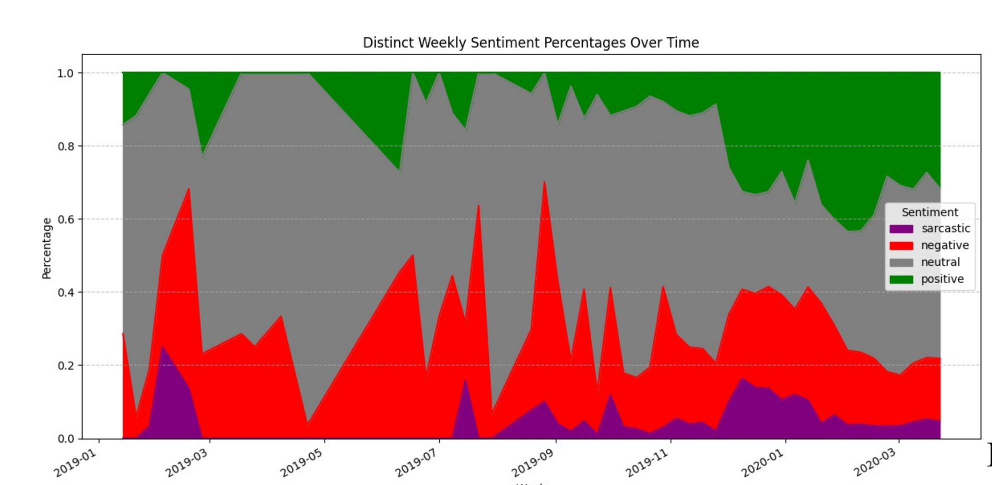
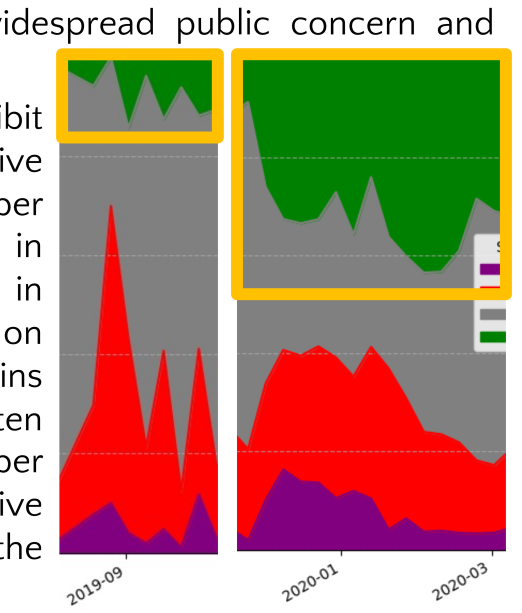


Fig. 7 b

Conclusion

Differentiation:

Figure 7 highlights two notable time periods: September 2019 and December 2019. In December 2019, there is a significant spike in negative posts, corresponding to the widely recognized outbreak of COVID-19. Similarly, in September 2019, a sudden increase in negative posts can be observed, attributable to the outbreak of African Swine Fever during this period. African Swine Fever, which primarily affects pigs, had a significant outbreak in South Korea. Given China's substantial number of South Korean restaurants and the interconnected food industry, this event triggered widespread public concern and panic on Weibo.



However, these two cases exhibit distinct patterns regarding positive sentiment posts. In the case of September 2019, there is no noticeable increase in positive posts alongside the surge in negative posts. The overall tone on Weibo during this period remains predominantly negative and often sarcastic. In contrast, during December 2019, a substantial rise in positive sentiment posts is observed following the outbreak of COVID-19 in China.

Several factors contribute to these differing patterns. Upon examining the data, we observed that crises with significant impacts on daily life tend to foster greater social cohesion and solidarity compared to less severe events. Two primary types of posts drive the rise in positive sentiment during such major crises: (1) government efforts to emphasize the effectiveness of disease control measures and (2) spontaneous expressions of encouragement and mutual aid among Weibo users.

In contrast, during the September 2019 African Swine Fever outbreak, the government showed little motivation to manage public speech about the disease, and its impact on the daily lives of ordinary citizens was minimal. As a result, public discussion was dominated by rumors, such as concerns about the contamination of supermarket pork. This lack of authoritative communication and reassurance exacerbated public panic, intensifying the negative sentiment on Weibo instead, compared to the outbreak of COVID-19.

Synchronization:

Another notable observation is the synchronization of dynamics among different types of sentiment. For instance, sarcastic sentiments and negative sentiments often exhibit synchronized patterns. A rise in negative posts is typically accompanied by a rise in sarcastic posts. This correlation is understandable, as sarcastic posts frequently convey negative opinions or criticisms.

Interestingly, synchronization also occurs between positive and sarcastic sentiments, particularly during abrupt surges in positive sentiment posts triggered by sudden events. In Figure 7, starting in December 2019, there is a marked increase in positive sentiment posts. During this period, sarcastic sentiment posts deviate from their usual alignment with negative sentiments and instead rise alongside the positive sentiments. A closer examination of these posts reveals that many sarcastic posts are directed at the surge in positive sentiments, often satirizing the shift in societal tone. Some sarcastic posts reflect arguments among Weibo users, while others criticize the Chinese government.

This phenomenon highlights that societal sentiments are not only shaped by real-world events but also influence one another. A surge in positive sentiments is often met with opposing voices, creating polarization within the online community. Such interactions underscore the complexity of sentiment dynamics in internet societies, where differing perspectives amplify societal divides.